MORDEN AIR QUALITY MONITORING SYSTEM

Embedded system

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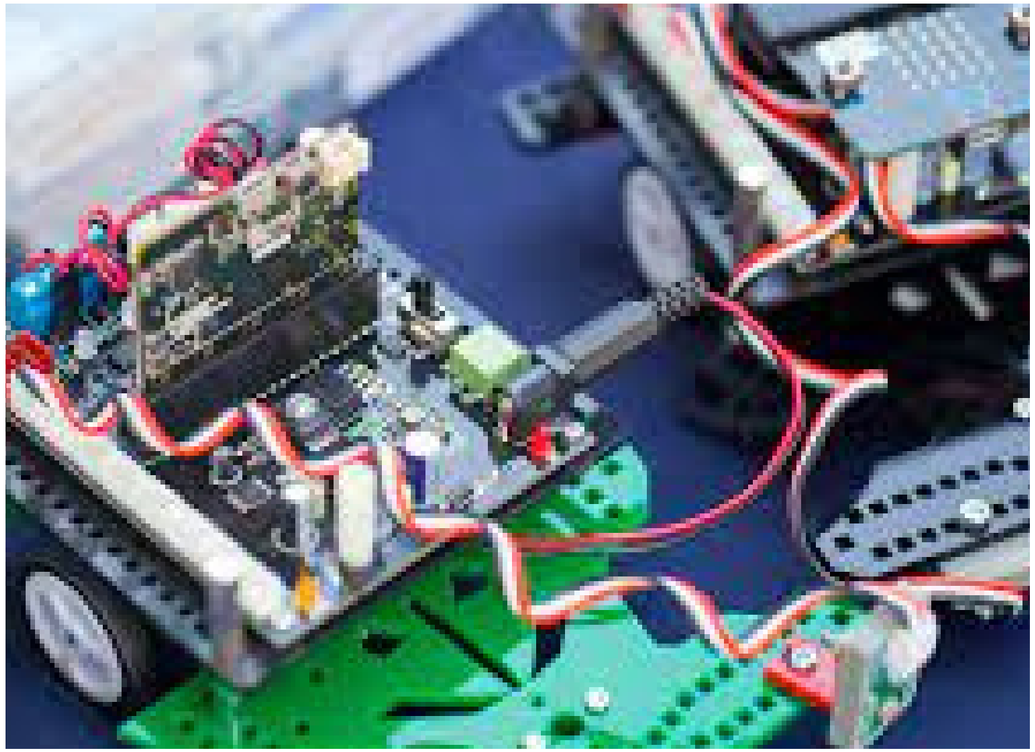
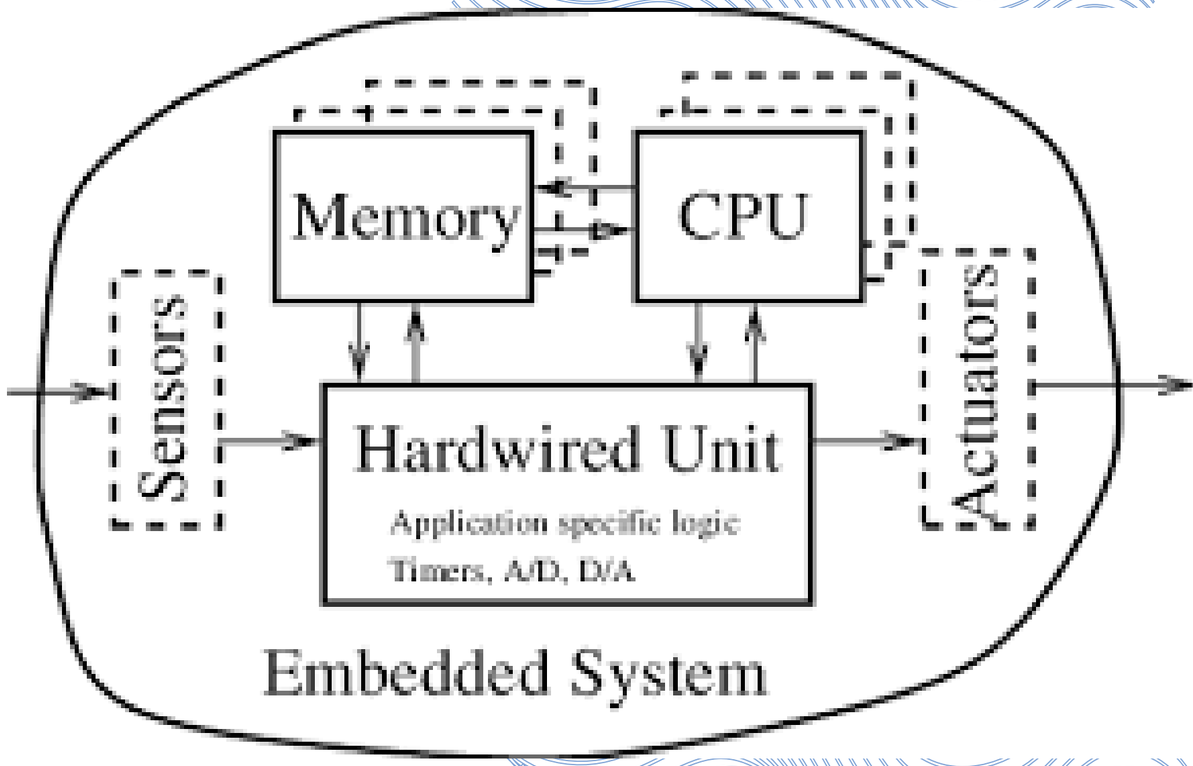


WHAT IS EMBEDDED SYSTEM

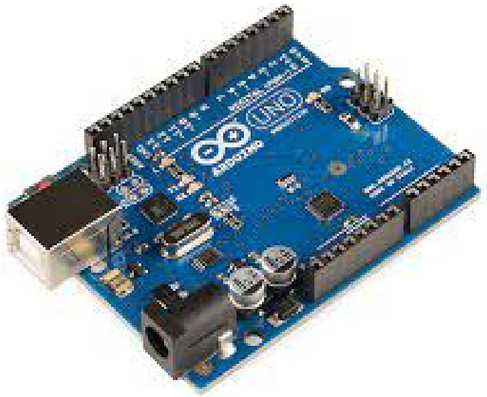
An embedded system is a microprocessor-based hardware system with software that is designed to perform a dedicated function, either as an independent system or as a part of a large system. At the core is an integrated circuit designed to carry out computation for real-time operations.



HOW IT IS IMPLEMENTED?

microcontroller- A microcontroller is a compact integrated circuit designed to govern a specific operation in an embedded system. A typical microcontroller includes a processor, memory and input/output (I/O) peripherals on a single chip. .

Develop the firmware or software that runs on the embedded system. This may include low-level device drivers, real-time operating systems (RTOS), middleware, application software, and user interfaces.



Arduino uno

STM32F401RE

ESP32

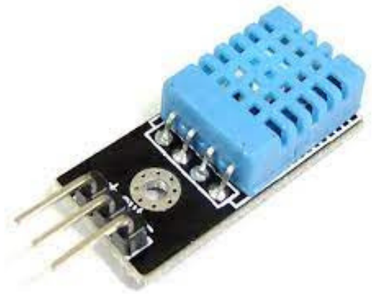
TI C2000

HOW IT IS IMPLEMENTED?

Sensors used in embedded systems are devices that detect changes in the physical environment and convert them into electrical signals that can be processed by the embedded system's hardware and software.nsors

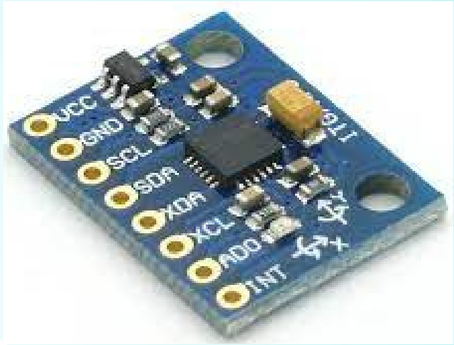
These sensors play a crucial role in enabling embedded systems to interact with and respond to their surroundings.

Sensors are fundamental building blocks in modern technology, enabling the collection of data for analysis, decision-making, and automation across a wide range of applications. As technology continues to advance, sensors are expected to become even more sophisticated, versatile, and integrated into everyday objects and systems.

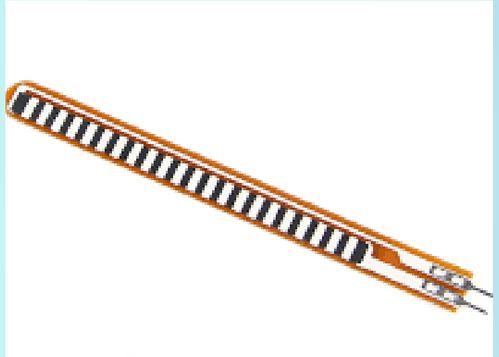


temperature sensor

IR sensor



mpu6050



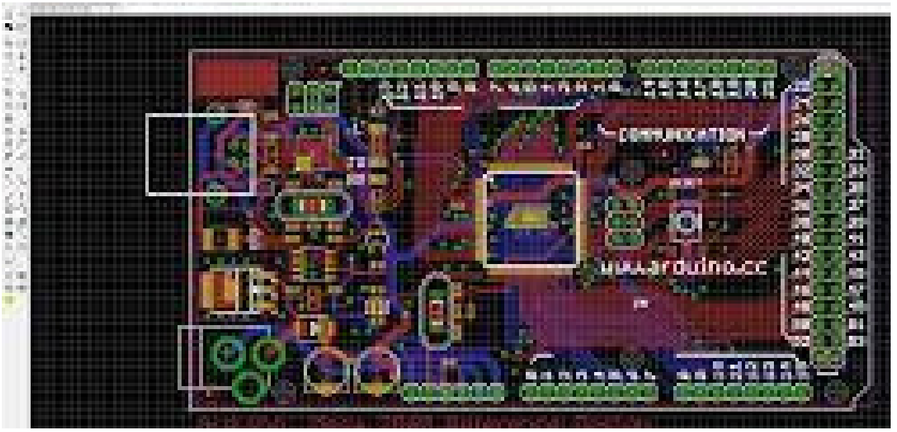
flex sensor

HOW IT IS IMPLEMENTED?

A PCB (Printed Circuit Board) is a fundamental component in modern electronics that provides a physical platform for mounting and interconnecting electronic components. It serves as a substrate for creating electrical circuits and facilitates the flow of signals and power between components. Here's a detailed explanation of what a PCB is and how it works:

Steps involved

- Schemetic Design
- PCB layout
- Fabrication
- Assembly
- Testing



Software - EAGLE,ALTIUM

COMMUNICATION PROTOCOL

UART (Universal Asynchronous Receiver/Transmitter)

- UART is a simple serial communication protocol commonly used for point-to-point communication between two devices.
- It uses two wires (TX and RX) for transmitting and receiving asynchronous serial data.

SPI (Serial Peripheral Interface)

- SPI is a synchronous serial communication protocol used for short-distance communication between a maste device and one or more slave devices.:
- It uses four wires: MOSI (Master Out Slave In), MISO (Master In Slave Out), SCK (Serial Clock), and SS (Slave Select).

I2C (Inter-Integrated Circuit)

- I2C is a synchronous serial communication protocol commonly used for communication between integrated circuits (ICs) on the same PCB.:
- It uses two wires: SDA (Serial Data) and SCL (Serial Clock), and supports multiple devices connected to the same bus.

WHERE IT IS IMPLEMENTED?

Embedded systems power Ocean Gate's exploration bots, navigating deep ocean depths, while Curiosity rover roams Mars, collecting invaluable data, all through sophisticated embedded technology.

RFID-enabled security locks leverage embedded systems, providing seamless access control with swift authentication, enhancing safety and convenience in modern environments.

WHERE IT IS IMPLEMENTED?

- Google Alexa
- Home automation with use of GSM communication technology

- Air quaiity monitering system- designed to provide real-time air quality data for informed decision-making. With features like portability, integration with smartphones, and cloud connectivity
- in medical field